



Shopify Product Tagger

A Confidence-Calibrated Multi-Agent LLM Pipeline

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The Problem

Merchants listing products on Shopify must manually assign taxonomy tags.

- **Slow** — tagging hundreds of SKUs by hand takes hours
- **Inconsistent** — different people apply different tags to similar products
- **Costly** — wrong tags hurt search ranking and recommendations

Goal: Build an automated pipeline that assigns the correct Shopify taxonomy tags to any product given its title and description.

Why Multi-Agent?

No single model handles all cases well.

Scenario	Challenge
Common products	Need speed and low cost
Ambiguous products	Need deeper language understanding
Out-of-distribution products	Need general world knowledge

Solution: Route each product to the most appropriate model based on confidence — fast when confident, powerful when not.

System Architecture

Product Title + Description



[Preprocessor]



[BERT Inference] ← always runs first



[Model Selector] ← confidence-based routing

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BERT LoRA- LLM
 Mistral API



[Validator Agent] ← checks tag coverage & confidence floor



Predicted Tags

Five Agents

Agent	Role
Preprocessor	Tokenizes, deduplicates, enforces min/max length
BERT Classifier	Fast inference + confidence estimation
Model Selector	Routes based on max confidence score
LLM API Tagger	Few-shot Claude Haiku / GPT-4o-mini fallback
Validator	Flags low-confidence or tag-sparse predictions

Routing Policy

```
if max_confidence ≥ 0.75:  
    route → BERT          # high confidence, fast path  
  
elif max_confidence ≥ 0.50:  
    route → LoRA-Mistral  # ambiguous, deeper model  
  
else:  
    route → LLM API       # out-of-distribution fallback
```

Cost increases left to right — the pipeline is designed to keep most traffic on the cheapest path.

Data

Source: McAuley Lab Amazon Reviews 2023 (HuggingFace Datasets)

Split	Products
Train	61,102
Val	7,638
Test	7,638
Total	76,378

- 21 Amazon categories → 30 Shopify taxonomy labels
- Avg labels per product: **1.08**
- Label coverage: **24 / 30** labels represented
- Streaming load — no disk overhead

Taxonomy — 30 Labels

electronics	computers	phones	audio
cameras	tv-video	clothing	shoes
jewelry	bags	beauty	health
sports	outdoors	home-kitchen	furniture
garden	tools-hardware	automotive	toys
baby	books	music	video-games
office	pet-supplies	food-grocery	arts-crafts
industrial	software		

Designed to map directly to Shopify's standard product taxonomy.

Phase 2 — BERT Baseline

Model: bert-base-uncased · 5 epochs · batch 32 · lr 2e-5

Metric	Score
Micro-F1	0.856
Macro-F1	0.652
Hamming Loss	0.011
Coverage Error	1.528

Inference latency (A100, single sample)

mean	median	p95	p99
15.3 ms	15.3 ms	15.7 ms	15.9 ms

BERT — Per-Label F1

Top performers

Label	F1
books	0.949
software	0.947
pet-supplies	0.946
food-grocery	0.942
video-games	0.939

Lowest — labels with sparse training data (computers, audio, clothing) scored 0.0 due to low representation across the 21 source categories.

Phase 3 — LoRA-Mistral-7B

Model: Mistral-7B · QLoRA 4-bit · r=16 · 3 epochs · batch 4 + grad accum 8

Metric	BERT	LoRA-Mistral
Micro-F1	0.856	0.868
Macro-F1	0.652	0.662
Hamming Loss	0.011	0.010
Coverage Error	1.528	1.478

LoRA-Mistral outperforms BERT on every metric — at a cost.

Latency Comparison

Model	Mean	p95	vs BERT
BERT	15.3 ms	15.7 ms	1×
LoRA-Mistral	262.7 ms	267.1 ms	17× slower

This tradeoff directly motivates the routing layer.

- BERT handles high-confidence products at ~15ms
- LoRA-Mistral only activates when confidence is ambiguous
- LLM API (~800ms + network) reserved for true out-of-distribution cases

Phase 5 — Live Routing Results

Routing breakdown on 7,638 test products:

Route	% of Traffic
BERT	100%
LoRA-Mistral	0%
LLM API	0%

Interpretation: BERT achieves max confidence ≥ 0.75 on virtually all in-distribution test products — the routing layer correctly identifies that LoRA and LLM API are not needed for this corpus.

Validator flagged: **0 / 497** products in demo set.

Example Routing Decisions

Product	Predicted Tags	Model	Confidence
Gotoh Modern Bridge for Tele	music	BERT	High
JOYIN 72 Pcs Animal Easter	toys	BERT	High
KZ ZS6 HiFi Quad Driver Earphones	music	BERT	High
Penny Rose Paper Dolls Bakery	toys, arts-crafts	BERT	High
Wearable4U H&K VP9 Airsoft	sports, outdoors	BERT	High

Interactive Demo

Gradio interface — live product tagger with routing visibility

- Input: product title + description
- Output: predicted tags with confidence scores
- Routing Decision panel: which model ran, why, flagged status

Built with `gradio.Interface` + `share=True` for public Colab URL.

Cost Analysis

At scale, routing prevents unnecessary LLM API spend.

Volume	BERT only	Full routing
1,000 products	~\$0	~\$0
10,000 products	~\$0	~\$0.01*
100,000 products	~\$0	~\$0.10*
1,000,000 products	~\$0	~\$1.00*

*LLM API cost only applies to the fraction routed to Claude Haiku (~\$0.80/M input tokens). For in-distribution products, cost is effectively zero.

Key Results Summary

	BERT	LoRA-Mistral
Micro-F1	0.856	0.868
Macro-F1	0.652	0.662
Latency	15 ms	263 ms
Cost/1M	~\$0	~\$0

The pipeline routes to the cheapest adequate model:

- Most products → BERT (fast, free, 85.6% F1)
- Ambiguous → LoRA-Mistral (+1.2% F1, 17× cost in latency)
- OOD → LLM API (general knowledge, network cost)

Limitations & Future Work

Current limitations

- 6 of 30 labels have zero training coverage (sparse categories)
- Routing thresholds tuned manually — could be learned
- LoRA-Mistral never activated on in-distribution test set

Next steps (SaaS roadmap)

- Shopify OAuth integration — tag products directly via API
- Merchant feedback loop — online learning from corrections
- Active learning to improve sparse label coverage
- REST API + merchant dashboard

Conclusion

A confidence-calibrated multi-agent pipeline that:

1. **Preprocesses** product text and deduplicates efficiently
2. **Classifies** with BERT at 85.6% micro-F1 and 15ms latency
3. **Routes** ambiguous cases to LoRA-Mistral or LLM API
4. **Validates** outputs before returning to the merchant
5. **Scales** cost-efficiently — most traffic never hits a paid API

GitHub: github.com/fdogarro/shopify-tagger

Video Presentation

Watch the full 7–15 minute demo and walkthrough:

[INSERT VIDEO URL HERE]

(YouTube / Zoom / Panopto)

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